A Gated Multimodal Architecture with Stochastic Weight Averaging for Directional Trading Decisions over Heterogeneous Financial Assets

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Abstract. We address the engineering of a computational architecture able to ingest and synchronize heterogeneous information sources for daily directional trading decisions: continuous price series, high-frequency unstructured text processed through a financial language model, and discrete low-frequency corporate filings. The central difficulty is not modelling each modality in isolation but fusing them coherently under severe asymmetries in frequency, dimensionality and availability. We propose a four-branch network that couples a patch-based temporal encoder, an attention pooling module over FinBERT news embeddings, a filing projection with temporal decay, and a per-day scalar encoder, integrated through a gated fusion layer that masks branches whose evidence is absent or inapplicable to the instrument. The architecture is validated on the FinMMEval Task 3 corpus (twelve assets spanning equities and cryptocurrencies). We contribute three engineering decisions: a per-asset volatility-adaptive labelling scheme, the decoupling of model optimization from decision-threshold optimization to curb overfitting on a temporally small validation set, and the use of Stochastic Weight Averaging (SWA). On the held-out test partition the validation-selected model attains a cumulative return of **23.3%**, an annualized Sharpe ratio of **4.21**, and a maximum drawdown of **4.45%**. We deliberately frame this as a validation-selected operating point rather than an expected value: across five seeds the test Sharpe varies widely (standard deviation ≈ 1.9), a direct consequence of the short evaluation window. We further document two negative results—SWA does not raise the across-seed mean, and prediction ensembles degraded out-of-sample performance—and argue that in this low-data regime the binding constraint is the experimental protocol rather than model capacity. All metrics are reported with explicit statistical caveats.

Keywords: Multimodal fusion · Deep learning · Financial NLP · FinBERT · Gated attention · Stochastic Weight Averaging · FinMMEval.

1 Introduction

Predicting the short-term direction of financial assets is a canonical testbed for multimodal learning. Market behaviour emerges from the interaction of price dynamics, the flow of textual information, and discrete corporate events, each differing in sampling frequency, granularity and semantics. The engineering challenge is to design an architecture that *synchronizes and weights* these signals without allowing high-cardinality modalities to dominate sparse but informative ones, and without forcing the model to integrate evidence that does not exist for a given instrument on a given day.

This work centres on the *architectural engineering* of multimodal fusion; the financial task serves as a controlled validation case rather than as the objective in itself. The same design concerns recur in any domain where continuous sensor streams, free text and discrete events must be combined.

Our contributions are: (i) a four-branch architecture with *gated fusion and availability masking*, where each modality can be differentiably silenced when its evidence is null or inapplicable; (ii) a *per-asset volatility-adaptive labelling* scheme that balances the class distribution across instruments with disparate volatility profiles; (iii) a training protocol that *separates model optimization from decision optimization*, fixing the activation threshold during training and tuning it only once at evaluation; (iv) the integration of *Stochastic Weight Averaging* (SWA) [1] as an implicit regularizer, with an ablation quantifying its effect; and (v) a transparent *negative result* on ensembling, which degraded out-of-sample performance in our low-data regime.

2 Related Work

Deep learning for financial forecasting has been surveyed extensively [12]. News-driven prediction was popularized by event-embedding models [10] and attention-based frameworks that weight news by relevance [11]. The introduction of domain-adapted language models, notably FinBERT [3,4], enabled robust sentiment and representation extraction from financial text. For the price stream, transformer encoders [5] have been adapted to time series; PatchTST [2] showed that segmenting a series into subseries-level patches yields strong, parameter-efficient forecasters.

Multimodal fusion has a well-established taxonomy [8]. Gated multimodal units [7] learn per-modality importance weights, a mechanism we extend with an explicit availability mask so that inapplicable modalities (e.g. filings for cryptocurrencies) are excluded by construction. On the optimization side, SWA [1] averages weights along the training trajectory to reach flatter optima that generalize better, and decoupled weight decay [6] stabilizes adaptive optimizers. For labelling, the difficulty of defining directional targets under non-stationary volatility is discussed in the financial machine-learning literature [14]; we adopt a per-asset percentile dead-zone instead of a global static threshold.

Table 1. Dataset summary after weekend consolidation and tensorization.

Property	Value
Assets	12 (10 equities, 2 crypto)
Date range	2025-08-01 to 2026-04-25
Raw daily records	2,446
Trading days per equity / crypto	191 / 268
Tensorized samples (train/val/test)	1,588 / 360 / 496
Unique trading dates (val / test)	40 / 53
Train class balance (S/H/B)	524 / 529 / 535
Scalar features selected	19 (of 69 candidates)
Sequence length L	30 days
Max news per day	8
News / filings coverage	all days / equities only

3 Problem Formulation and Data

3.1 Multimodal formulation

For an asset at day t we observe $\mathcal{X}_t = \{\mathbf{S}_t,\ \mathcal{N}_t,\ \mathcal{E}_t\}$, where $\mathbf{S}_t \in \mathbb{R}^{L\times d_s}$ is a window of L days of scalar technical features, $\mathcal{N}_t = \{n_t^{(1)}, \ldots, n_t^{(k_t)}\}$ is a variable-size set of news headlines, and $\mathcal{E}_t$ encodes corporate filings (10-K, 10-Q). The goal is a function $f_\theta : \mathcal{X}_t \mapsto y_t \in \{\textsc{sell}, \textsc{hold}, \textsc{buy}\}$ producing one daily recommendation per asset.

3.2 Corpus

The corpus is provided by the *FinMMEval Task 3* framework and is built from the publicly released `daily_news` collection accessed from the Hugging Face Hub (`TheFinAI/daily_news`). It comprises twelve assets—ten US equities (AAPL, ADBE, AMZN, BMRN, GOOGL, META, MRNA, MSFT, NVDA, TSLA) and two cryptocurrencies (BTC, ETH)—over the period 2025-08-01 to 2026-04-25. Each daily record contains the closing price, a momentum tag, the set of news for the day, and optional filings. Table 1 summarizes the dataset after preprocessing.

For equities, weekend and holiday records (no trading) are removed and their news are carried forward to the next trading day with an exponential temporal decay $w = \exp(-\lambda\,\Delta_{\text{days}})$, $\lambda = 0.18$, so that Friday news reaching Monday retain weight ≈ 0.70. Cryptocurrencies trade seven days a week and require no consolidation. Temporal splits are performed per asset (65%/15%/20%) and concatenated, preserving chronological order and preventing look-ahead leakage.

4 Methodology

4.1 Feature engineering and dynamic labelling

A single feature-engineering module is shared between offline training and the serving runtime, guaranteeing reproducibility. From roughly sixty candidate indicators (returns at multiple horizons, realized volatilities, RSI, MACD, Bollinger position, trend slopes, calendar dummies, news counts, filing-decay features and cross-asset proxies) we select 19 features through the combination of Spearman correlation, mutual information, LightGBM [9] gain under a purged walk-forward cross-validation with a five-day embargo, and a variance inflation factor filter to remove collinear variables.

Target construction. Let $r_{t,a} = (p_{t+1,a} - p_{t,a})/p_{t,a}$ be the one-day-ahead realized return. A per-asset *dead-zone* threshold τ_a is estimated as the q-th percentile of $|r|$ over an initial in-sample window W_a of each asset (the first 70% of its series):

$$\tau_a = \mathrm{P}_q\big(\{|r_{s,a}| : s \in W_a\}\big), \qquad q = 33, \tag{1}$$

and the categorical label is

$$y_{t,a} = \begin{cases} \text{BUY}, & r_{t,a} > +\tau_a, \\ \text{SELL}, & r_{t,a} < -\tau_a, \\ \text{HOLD}, & |r_{t,a}| \le \tau_a. \end{cases} \tag{2}$$

A static global threshold under-represents the active class for low-volatility instruments and over-represents noise for volatile ones. The per-asset percentile produces thresholds ranging from 0.18% (MSFT) to 0.86% (ETH), which track each instrument's volatility and keep the three classes approximately balanced (Table 1). An auxiliary regression target, the return normalized by a 20-day rolling realized volatility (lagged one day to avoid leakage), supervises a secondary head.

On the estimation window and look-ahead. The window W_a (first 70%) is slightly wider than the 65% training split, so it overlaps the first few validation days; it never reaches the test partition. This is deliberate and has no measurable effect on the reported results, for two reasons. First, τ_a is a single per-asset *magnitude* derived from $|r|$ (no directional or future information) and, empirically, recomputing it on the strict 65% training split leaves *all* 360 validation labels unchanged and alters only 2 of 496 test labels (0.4%); model selection, which is performed on validation, is therefore invariant. Second, the headline financial metrics (CR, SR, MD, DV, AV) are computed from realized returns and the model's discrete signals and are *independent of* τ_a, which enters only the auxiliary classification metrics (accuracy and macro-F1). The reported test performance is thus free of look-ahead.

4.2 Architecture

The model (Fig. 1) has four modality branches whose d-dimensional outputs are integrated by a gated fusion layer. Identity information—an 8-dimensional learned symbol embedding concatenated with an asset-type flag—is appended before the output heads, so the network always knows which instrument it is scoring.

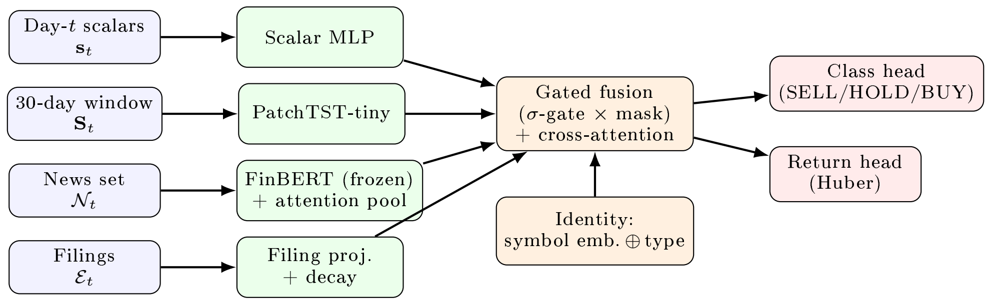

Fig. 1. Multimodal architecture. Four modality branches feed a gated fusion layer with availability masking; learned identity features are appended before the classification and regression heads. FinBERT is frozen; only 115K parameters are trained.

Branches. (1) A scalar MLP encodes the day-t feature vector. (2) A compact PatchTST [2] encoder segments the 30-day window into non-overlapping patches of length five and applies a two-layer transformer with mean pooling. (3) News are embedded with a frozen FinBERT (`finbert-tone`) [3]; the day's headlines are aggregated by an attention pooling whose query is the concatenation of the symbol embedding and the asset-type flag, with attention logits additively biased by the temporal decay weight. (4) A filing projection encodes the running filing embedding after applying a 45-day half-life decay.

4.3 Gated fusion with availability masking

Let $\mathbf{h}_m \in \mathbb{R}^d$ be the representation of branch m and $a_m \in \{0, 1\}$ its availability indicator ($a_{\text{filing}} = 0$ for crypto; $a_{\text{news}} = 0$ on days without news). A gate is computed from the concatenation of all branches and modulated by availability:

$$g_m = \sigma\big(W_g\,[\mathbf{h}_1 \| \cdots \| \mathbf{h}_4]\big)_m \cdot a_m. \tag{3}$$

Gated representations $g_m \mathbf{h}_m$ are refined by light cross-attention across the four branches (treated as tokens) and pooled with weights proportional to g_m. This design prevents the network from being forced to integrate non-existent evidence.

4.4 Loss, optimization and stabilization

The objective is multi-task, $\mathcal{L} = \alpha\,\mathcal{L}_{\text{CE}} + (1 - \alpha)\,\mathcal{L}_{\text{Huber}}$, with $\alpha = 0.8$, class-balanced cross-entropy with label smoothing [13] of 0.10, and a Huber loss on the volatility-normalized return. We optimize with AdamW [6] under a five-epoch warmup and cosine decay, with gradient clipping and three temporal data augmentations (Gaussian feature noise, news dropout, window time-shift).

Decoupling decision from optimization. Converting class probabilities into a discrete action requires an activation threshold θ. Optimizing θ jointly with the weights—selecting, each epoch, the value that maximizes a financial metric on validation—induces overfitting when the validation set has *few unique trading dates* (here, 40). The annualized Sharpe ratio over so few days has large sampling variance, and a fine threshold sweep exploits accidents of the particular realization. We therefore (i) hold $\theta = 0.50$ *fixed* during training and select the model by a combined score $0.5\,\tanh(\mathrm{SR}/3) + 0.5\,\mathrm{F1}$ that is robust to the threshold, and (ii) tune θ *once* on validation after training, by grid search in $[0.34, 0.70]$, transferring the chosen value to the test set.

Stochastic Weight Averaging. SWA [1] averages the weights visited in the final training phase, $\bar{\theta}_{\mathrm{SWA}} = \frac{1}{K}\sum_k \theta_k$, approximating the centre of a flat region of the loss surface. Flat minima are less sensitive to distribution shift between validation and test, which is the dominant failure mode in this low-data regime. We start averaging at epoch 30 with a reduced constant learning rate and recompute normalization statistics with one post-training pass over the training set. The averaged checkpoint is evaluated as an independent model.

5 Experimental Setup

Table 2 lists the hyperparameters of the deployed configuration. The model has 115,016 trainable parameters (the frozen FinBERT encoder is excluded). We report portfolio-level metrics computed on daily returns aggregated across assets with equal weight over active (non-HOLD) positions, annualizing with a factor of 252: Cumulative Return (CR), annualized Sharpe Ratio (SR), Maximum Drawdown (MD), Daily Volatility (DV) and Annualized Volatility (AV) [15]. The threshold selected on validation is transferred unchanged to the test set.

6 Results

6.1 Main results

We compare four checkpoints, all sharing the architecture of Section 4.2 and the same corpus and splits. The *prior baseline* is an earlier checkpoint of the identical architecture (d=32, two PatchTST layers, dropout 0.35) trained under our previous protocol, which selected the model on a noisy simulated Sharpe and tuned the decision threshold *during* training—the practice this paper argues against (Section 4.4). The *base* and *proposed* models use the present protocol (fixed training threshold, validation-only threshold tuning); "base" is the non-averaged checkpoint and "proposed" applies SWA to it. We stress that "proposed" denotes the *validation-selected operating point* of our protocol, not a method claimed to be universally superior; the across-seed analysis (Section 6.3) qualifies this directly.

Table 3 reports test-set performance of the validation-selected checkpoints. The proposed SWA model attains the highest cumulative return (**23.3%**) and Sharpe ratio (**4.21**) and the lowest drawdown (**4.45%**) among the compared

Table 2. Hyperparameters of the deployed configuration.

Component	Hyperparameter	Value
Architecture	Fusion dimension d	32
	PatchTST layers / patch length	2 / 5
	Symbol embedding dim	8
	Trainable parameters	115,016
Optimization	Optimizer	AdamW
	Learning rate	1×10^{-3}
	Weight decay	1×10^{-2}
	Batch size	32
	Dropout	0.35
	Max epochs / warmup / patience	80 / 5 / 20
Loss	α (CE weight)	0.80
	Label smoothing	0.10
	Class weights	balanced
Augmentation	Gaussian noise σ	0.03
	News dropout prob.	0.30
	Time shift	± 2
SWA	Start epoch	30
	SWA learning rate	5% of base

checkpoints; the non-averaged model from the same seed reaches a Sharpe of 2.88, and a five-seed ensemble (last row) *underperforms* both (Section 6.3). These are operating points selected on validation; Section 6.3 reports the across-seed distribution and should be read alongside this table. Figure 2 shows the out-of-sample equity curves: the SWA model is more selective (it opens fewer positions at the higher tuned threshold) and produces a smoother trajectory in the second half of the test window.

6.2 Ablation: effect of SWA

Table 4 isolates SWA on the deployed (validation-selected) seed. For this checkpoint, averaging the final-phase weights improves the Sharpe ratio from 2.88 to 4.21 and reduces the maximum drawdown from 5.2% to 4.5%. We caution against over-reading this within-seed gain: as shown in Section 6.3, SWA helped three of five seeds and hurt two, and did not improve the across-seed mean. The honest reading is therefore that SWA yielded the single best validation-selected model in our runs, not that it uniformly improves generalization on this corpus. We also note that the SWA model attains a lower macro-F1 (0.27 vs. 0.40): at the higher tuned threshold it trades sparsely and selectively, raising risk-adjusted return while lowering nominal classification agreement—a reminder that accuracy and risk-adjusted profit are distinct objectives.

Table 3. Test-set performance. Threshold θ tuned once on validation and transferred. Best value per column in bold.

Model	θ	CR	SR	MD	DV	AV
Prior baseline	0.56	+0.085	2.21	0.074	0.012	0.183
Prior baseline + SWA	0.40	+0.175	3.32	0.065	0.015	0.239
Base (no SWA)	0.50	+0.168	2.88	0.052	0.017	0.268
Proposed (SWA)	0.70	**+0.233**	**4.21**	**0.045**	0.015	0.244
5-seed ensemble	0.52	+0.029	0.81	0.104	0.012	0.189

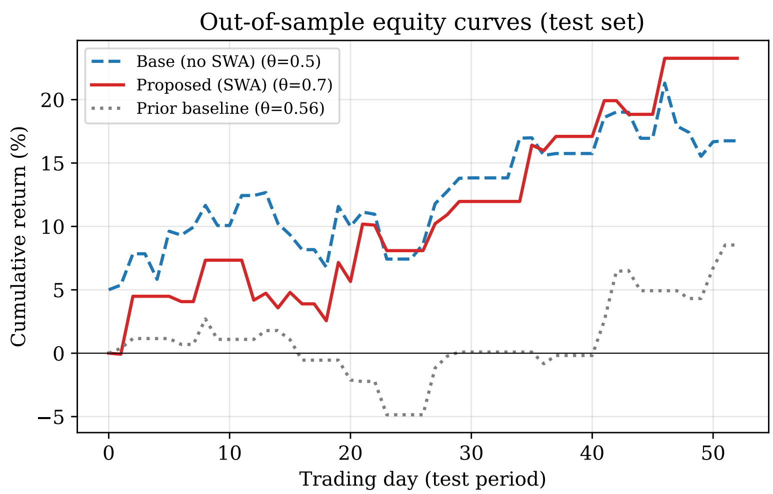

Fig. 2. Out-of-sample cumulative return over the 53-day test window for the prior baseline, the non-averaged model and the proposed SWA model.

6.3 Seed robustness

Because the validation set spans few unique dates, single-seed test estimates carry substantial variance. Table 5 reports the distribution over five seeds of the deployed configuration; in every case the operating checkpoint is chosen by the validation criterion before being measured on test. The variance is large for both variants (test-SR standard deviation $\approx$ 1.8–2.0), and three observations follow. First, the headline figure (SR 4.21) corresponds to the seed with the highest validation Sharpe (5.54), i.e. it is the *validation-selected operating point*, not an expected value. Second, SWA improved three of the five seeds but degraded the other two, so it does *not* raise the across-seed mean in this regime; its concrete benefit here is to produce the single best validation-selected checkpoint. Third, the wide spread is itself the central empirical message: with 53 test dates, point estimates of risk-adjusted return are statistically fragile and must be read as such.

Table 4. SWA ablation on the deployed (validation-selected) seed, test set. Within-seed deltas; see Section 6.3 for the across-seed picture.

Configuration	SR	CR	MD	macro-F1	accuracy
Base (no SWA)	2.88	+0.168	0.052	0.395	0.438
+ SWA	**4.21**	**+0.233**	**0.045**	0.269	0.363
Δ relative	+46%	+39%	−15%	−32%	−17%

Table 5. Per-seed validation and test Sharpe for the deployed configuration; the operating threshold θ is tuned on validation per seed. The deployed model is the SWA checkpoint of the validation-selected seed (seed 2, highest SWA validation Sharpe, 5.54); the *base* row reports the same seed's non-averaged checkpoint for a controlled within-seed ablation (both shown in bold). The last two columns give the across-seed mean and sample standard deviation of the test Sharpe.

Variant	metric	s0	s1	s2	s3	s4	mean	std
Base	val SR	6.08	5.53	**3.57**	6.27	5.57	—	—
	test SR	−0.19	3.28	**2.88**	2.76	−0.22	+1.70	1.75
SWA	val SR	2.69	4.05	**5.54**	4.31	2.44	—	—
	test SR	0.08	−0.17	**4.21**	−0.49	0.09	+0.75	1.95

6.4 Per-asset analysis

Figure 3 decomposes the SWA model by instrument. The strategy is profitable on seven of twelve assets (notably TSLA, ETH and MRNA, with per-asset Sharpe above 2.5) and loses on five (ADBE and BMRN being the worst). This heterogeneity indicates that a single shared model does not transfer uniformly across instruments and motivates instrument-aware calibration as future work.

6.5 Negative result: ensembling

Contrary to the usual variance-reduction expectation, both a five-seed ensemble and a top-k configuration ensemble *degraded* out-of-sample performance. The five-seed probability-averaging ensemble of the deployed configuration reached a test Sharpe of +0.81 (CR +2.9%), far below the validation-selected single model and below the base across-seed mean. Averaging weights systematically wrong members equally with good ones; given the large seed variance documented in Section 6.3, a subset of members collapses toward the majority HOLD prior and the average inherits that collapse. We conclude that, with a validation set of low temporal cardinality, member filtering is a prerequisite for ensembling, and we adopt a single-model-with-SWA strategy instead.

6.6 Training dynamics and confusion structure

Figure 4 shows that the training loss decreases smoothly while the validation Sharpe (at fixed threshold) is highly volatile—direct evidence for the small-

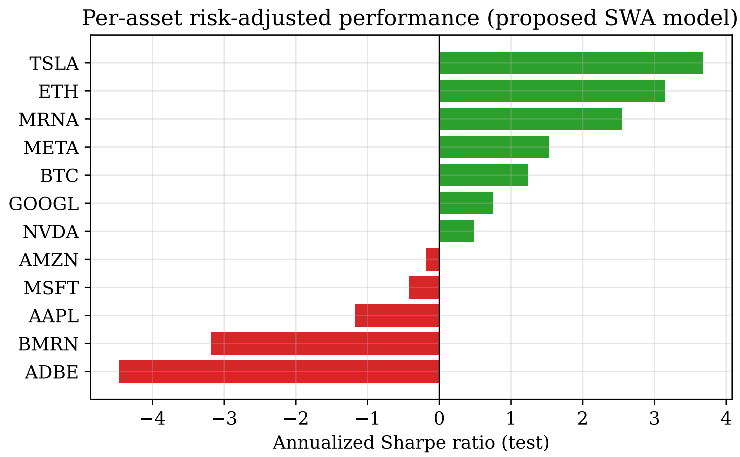

Fig. 3. Per-asset annualized Sharpe ratio on the test set for the proposed SWA model. Green: profitable; red: unprofitable.

sample argument motivating our threshold policy. Validation macro-F1 is comparatively stable, which is why it anchors the selection score. Figure 5 shows the test confusion matrix of the SWA model; the high tuned threshold concentrates predictions on HOLD, with the active classes used selectively.

7 Discussion and Limitations

The improvements stem from pipeline-level engineering—volatility-adaptive labelling, the separation of model and decision optimization, and SWA—rather than from radical architectural novelty, suggesting that in multimodal problems with limited data the binding constraint is often the experimental protocol rather than model capacity.

Several limitations must be stated explicitly. *(i) Short evaluation horizon.* The test set spans only 53 unique trading dates (≈2.5 months); an annualized Sharpe obtained by scaling a short-window mean-to-standard deviation ratio by $\sqrt{252}$ has wide confidence intervals, and we did not compute bootstrap intervals. The headline figure should therefore be read as indicative, not definitive. *(ii) Small validation set.* With 40 unique dates, the single threshold tuned on validation is itself a high-variance estimate. *(iii) Seed sensitivity.* Performance varies strongly across seeds (test-SR standard deviation $\approx$ 1.8–2.0); the deployed checkpoint was chosen by validation score, but the across-seed mean is far below the deployed value and SWA does not improve that mean (Table 5). The contribution of this paper is therefore better stated as a study of protocol under data scarcity than as a claim of a robustly superior model. *(iv) No transaction costs or slippage* are modelled, and the equal-weight active-position portfolio is

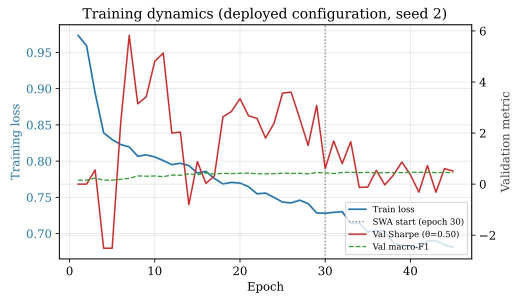

Fig. 4. Training loss (left axis) and validation metrics (right axis). The dotted line marks the SWA start epoch.

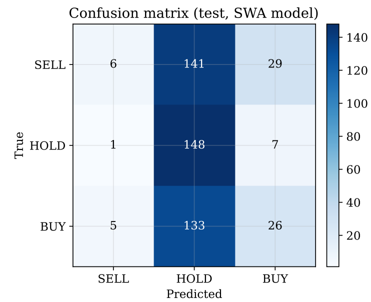

Fig. 5. Test confusion matrix (SWA model).

a simplification. *(v) Data-source provenance.* The competition-venue metadata could not be independently verified and is reported as provided.

8 Conclusion

We presented a gated multimodal architecture that fuses price series, FinBERT news embeddings and corporate filings through availability-masked gating, and we validated it on the FinMMEval Task 3 corpus. The validation-selected model, combining per-asset adaptive labelling, decoupled threshold optimization and Stochastic Weight Averaging, achieved a test cumulative return of 23.3%, a Sharpe ratio of 4.21 and a maximum drawdown of 4.45%. We reported these as validation-selected operating points and showed, through a seed study, that the across-seed distribution is wide and that neither SWA nor ensembling robustly improves the mean in this short-horizon regime—results we present as a methodological caution rather than as a universally superior model. Future work includes bootstrap confidence intervals over a longer test horizon, filtered ensembling, instrument-aware calibration, and parameter-efficient fine-tuning of the text encoder.

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References

1. Izmailov, P., Podoprikhin, D., Garipov, T., Vetrov, D., Wilson, A.G.: Averaging Weights Leads to Wider Optima and Better Generalization. In: Proc. Uncertainty in Artificial Intelligence (UAI), pp. 876–885 (2018)
2. Nie, Y., Nguyen, N.H., Sinthong, P., Kalagnanam, J.: A Time Series is Worth 64 Words: Long-term Forecasting with Transformers. In: Int. Conf. on Learning Representations (ICLR) (2023)

3. Yang, Y., Uy, M.C.S., Huang, A.: FinBERT: A Pretrained Language Model for Financial Communications. arXiv:2006.08097 (2020)
4. Araci, D.: FinBERT: Financial Sentiment Analysis with Pre-trained Language Models. arXiv:1908.10063 (2019)
5. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., Polosukhin, I.: Attention is All You Need. In: Advances in Neural Information Processing Systems (NeurIPS), pp. 5998–6008 (2017)
6. Loshchilov, I., Hutter, F.: Decoupled Weight Decay Regularization. In: Int. Conf. on Learning Representations (ICLR) (2019)
7. Arevalo, J., Solorio, T., Montes-y-Gómez, M., González, F.A.: Gated Multimodal Units for Information Fusion. In: ICLR Workshop Track (2017). arXiv:1702.01992
8. Baltrušaitis, T., Ahuja, C., Morency, L.-P.: Multimodal Machine Learning: A Survey and Taxonomy. IEEE Trans. Pattern Anal. Mach. Intell. **41**(2), 423–443 (2019)
9. Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., Liu, T.-Y.: LightGBM: A Highly Efficient Gradient Boosting Decision Tree. In: Advances in Neural Information Processing Systems (NeurIPS), pp. 3146–3154 (2017)
10. Ding, X., Zhang, Y., Liu, T., Duan, J.: Deep Learning for Event-Driven Stock Prediction. In: Proc. Int. Joint Conf. on Artificial Intelligence (IJCAI), pp. 2327–2333 (2015)
11. Hu, Z., Liu, W., Bian, J., Liu, X., Liu, T.-Y.: Listening to Chaotic Whispers: A Deep Learning Framework for News-oriented Stock Trend Prediction. In: Proc. ACM Int. Conf. on Web Search and Data Mining (WSDM), pp. 261–269 (2018)
12. Sezer, O.B., Gudelek, M.U., Ozbayoglu, A.M.: Financial Time Series Forecasting with Deep Learning: A Systematic Literature Review: 2005–2019. Applied Soft Computing **90**, 106181 (2020)
13. Szegedy, C., Vanhoucke, V., Ioffe, S., Shlens, J., Wojna, Z.: Rethinking the Inception Architecture for Computer Vision. In: Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR), pp. 2818–2826 (2016)
14. López de Prado, M.: Advances in Financial Machine Learning. Wiley, Hoboken, NJ (2018)
15. Sharpe, W.F.: The Sharpe Ratio. Journal of Portfolio Management **21**(1), 49–58 (1994)